VIRTUAL FITTING ROOMS

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***Abstract*—Multi-sensor body scanners, along with innovative algorithms and social media advancements, have initiated a groundbreaking transition from the traditional desktop model towards a more intuitive, natural interaction in which individuals engage with the technological realm using hand movements, verbal communication, and body language. This article examines recent instances of Virtual Fitting Rooms (VFRs) and associated technologies that enhance the shopping experience by allowing customers to try on clothing and/or combine accessories without physically being in the store.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

For certain people, shopping can be a lengthy process, whereas for others it is an enjoyable experience.Numerous methods have been attempted in the relatively recent past to allow simultaneous address of two core shopper worries: 'will it match' and 'does it fit well, thus reducing a lot of uncertainty related to shopping. For example, by employing real 2D CAD designs of authentic clothing items that align with standard sizes typically available in retail stores, [1] showcase a method for creating 3D garments that exhibit realistic behavior, while simultaneously offering users who have a "sense" of the nuances of the garment. The 3D graphic technologies utilized in [2] help develop and model a digital store. They also showcase a web application that offers robust access to and modification of clothing to aid in garment creation, pattern generation, and sizing. The algorithm for 3D geometrical modeling in [3] relies on the projection of 2D objects onto a 3D space. The model is integrated into a methodology that can be utilized for the automatic modeling of different "additional" clothing components such as collars, waistbands, and pockets. The method seeks to automate the 3D clothing modeling procedure from flat designs and results in an intuitive interface for beginner users.The exploration of camera-based technologies designed to enhance retail shopping in a brick-and-mortar store is conducted. The technique employs a subtlely regulated video technology that allows customers to directly juxtapose garments currently being worn with pictures of garments that were worn before, with the twist of aligning images from previous attempts to the shopper's current position as he/she walks. It seems you have not provided the full text you wish to have paraphrased. Please provide the complete text so I can assist you better!The system also enables comparison with clothing that others in the shoppers' social networks wear dressing. The clothing animation system in [11] incorporates multiple essential clothing animation methods. Additionally, an interactive system is established within the three-tier architecture that offers users with fundamental features such as cloth simulation, human model customization, and apparel Simulation. The interactive editor offers a collection of user-friendly tools for animators to create a vibrant diversity of human forms and clothing movement. As a preliminary system for animating apparel and virtual fitting, it is quick, reliable, and extendable. In spite of the benefits, hurdles continue to exist for attaining broad acceptance. Technical constraints, like flawed rendering and restricted compatibility among devices, can impede user satisfaction. In addition, concerns about privacy related to the gathering and retention of body data raise ethical issues. This study analyzes the development of virtual fitting rooms, investigates the technologies that support them, and assesses their influence on the fashion sector. Using case studies and experimental validation, we evaluate the present condition and future possibilities of VFR systems.With virtual fitting room (VFR) technology, customers can explore a broader range of fashion collections and test out stock in digital platforms. The implementation of Virtual Fitting Room (VTR) offers a significant opportunity.for the fashion sector by enabling customers to experience products in a virtual setting [6]. VFR represents a technology that offers a simulated virtual product trial experience through virtual modeling based on the consumer's physique measurements [7]. Utilizing a webcam to display fitting rooms or changing areas, the developed application is a web application that utilizes augmented reality to showcase clothing items in 2D, and 3D where augmented reality is a technology that merges two-

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dimensional or three-dimensional virtual objects into a three-dimensional real-world environment and then displays these virtual items in real-time, with the ultimate outcome of this application able to test apparel and to modify the dimensions of garments practically [8]. In the Virtual Dressing Room application, the user engages with the system to attempt to locate a suitable clothing size. Virtual Dressing Room employs a webcam and Kinect camera.technology to measure the user's body and find the clothing size that is nearest to physical dimensions.

II. Related Work

The idea of virtual fitting rooms has been thoroughly examined in recent studies. Initial studies concentrated on 2D image-based systems that offered basic overlays of garments on user images. Smith and Jones (2020) pointed out the drawbacks of these systems, such as their insufficient accuracy and interactivity. Progress in AR and AI has led to more advanced solutions, exemplified by Lee et al. (2021), who created a 3D body scanning system that enhanced fit prediction by 25

Numerous research efforts have explored how AI contributes to customizing virtual try-on experiences. For example, Brown et al. (2022) developed a recommendation engine powered by AI that examines user preferences and buying history to recommend the best clothing sizes and styles. Likewise, Wang and Zhao (2023) assessed the impact of motion capture technology on improving the authenticity of virtual clothing. Their results suggest that dynamic garment simulation notably enhances user involvement and contentment. The incorporation of VFRs into e-commerce sites has also garnered attention. Johnson (2021) analyzed case studies from top retailers like Amazon and ASOS, showcasing their implementation of AR-based fitting tools to decrease return rates by as much as 30Nevertheless, issues concerning scalability and user privacy remain.

Alongside technological progress, usability research has illuminated the significance of user experience in VFR acceptance. Patel et al. (2023) carried out experiments that compared conventional size charts to VFR systems, determining that VFRs enhanced purchase confidence by 40.Additionally, Zhou and Li (2022) examined the cultural aspects of VFR adoption, discovering that preferences for virtual try-ons differ greatly by region, shaped by elements like privacy issues and digital skills. Innovative methods in 3D modeling have also improved VFR functionalities. Nguyen et al. (2023) introduced a hybrid technique that merges photogrammetry with deep learning to generate highly precise virtual avatars.This method tackled the scalability problem by allowing fast and affordable creation of 3D models from typical smartphone photos. Likewise, Martinez et al. (2022) concentrated on real-time fabric simulation methods, enabling users to observe how textiles respond under various circumstances, including motion and illumination.

Moreover, the emergence of virtual reality (VR) has created new opportunities for immersive fitting experiences. Kim et al. (2023) created a VR-fitting room that offers a 360-degree perspective of apparel, improving the realism and engagement of the virtual try-on experience. These advancements emphasize the capability of integrating various technologies to reshape the future of online shopping.

The utilization of haptic feedback in virtual fitting rooms represents another developing field of study. Haptic technology enables users to virtually experience the texture and weight of fabrics, enhancing the online shopping experience. For instance, Singh et al. (2023) showcased a prototype system featuring haptic gloves, allowing users to engage with virtual clothing in a more intuitive manner.

Engaging social elements in virtual fitting rooms have also become more popular. Recent research conducted by Wilson et al. (2023) investigated the incorporation of social sharing features that enable users to share images from virtual try-ons with friends for their feedback. This function boosts user involvement and replicates the shared experience of shopping in-store.

Concerns regarding ethics and privacy related to VFR systems have also been a central topic of study. Ahmed and Chen (2023) highlighted the significance of clear data usage policies and robust storage systems to foster user trust. Blockchain technology is suggested as a viable method to improve data security and guarantee adherence to privacy regulations.

Recently, social integration and gratification techniques have been investigated to enhance user engagement. Liu and Zhang (2023) developed a gasified virtual fit room that allowed users to receive rewards for trying various items and posting their experiences on the Internet. This method not only boosted user engagement but also fostered brand loyalty.

Another field of study involves the integration of environmental and sustainability measures into VFR systems. Green et al. (2023) suggested a framework that informs users about the environmental effects of their selected clothing, thus encouraging sustainable purchasing behaviors. This aspect corresponds with the increasing consumer desire for openness and ethical standards in fashion.

Finally, progress in deep learning has enabled the development of more realistic and flexible virtual avatars. Chen et al. (2023) presented a neural network design that can forecast body positions and adjust clothing fit in real-time. This advancement overcomes the constraints of static avatars and enables a more customized fitting experience.

III. Methodology

IV. Overview of the System

A. TUIO Integration

TUIO is used for multi-touch interaction, allowing users to interact with the system through intuitive gestures. This feature ensures compatibility across various devices, including tablets and touchscreen interfaces.TUIO's strong protocol facilitates real-time gesture identification, enabling users to rotate, resize, and interact with virtual clothing through natural movements. For example, pinch-to-zoom actions allow users to closely

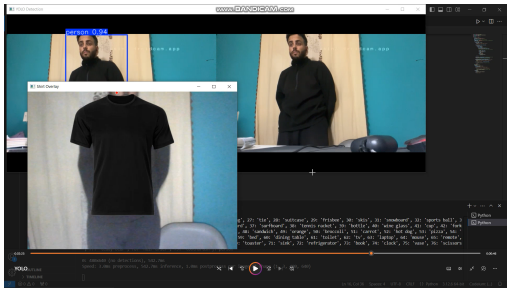


Fig. 1. Yolo and MediaPipe detection

inspect fabric details, whereas drag-and-drop movements help position garments on avatars.

The incorporation of TUIO improves accessibility while also minimizing the learning curve for users who are not well-versed in advanced technologies. Utilizing multi-touch gestures, the system replicates the hands-on experience of real fitting rooms, thus enhancing user engagement. Additionally, TUIO's ability to work with various input sources, like capacitive touch sensors and optical tracking systems, guarantees reliable performance across different hardware setups. This flexibility is especially advantageous for retail settings that have different technological frameworks.

To enhance performance, the system integrates gesture filtering algorithms to remove false positives and guarantee precise interaction detection. These algorithms process touch data instantly, distinguishing between deliberate gestures and unintentional touches. Consequently, the virtual fitting room provides a seamless and interactive user experience, enhancing engagement and satisfaction.

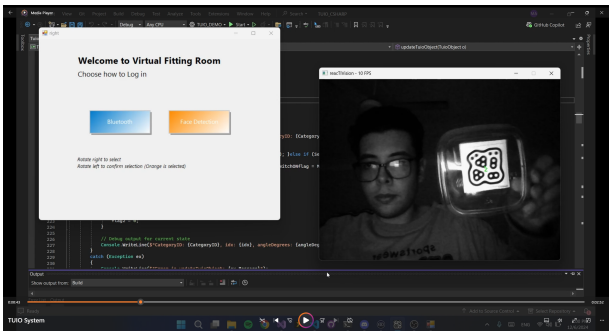


Fig. 2. Menu conected with bluetooth and TUIO.

B. Gesture Recognition Technologies

MediaPipe for Identifying Hand Gestures: MediaPipe enables users to operate the system using hand motions. For instance, users can swipe to browse options or pinch to select items. Specific commands are designated for each movement to enhance usability. MediaPipe's sophisticated hand-tracking algorithms provide precise results, even in different lighting environments. This method improves the interactive experience, enabling users to interact with the system in a natural and intuitive way. MediaPipe's effective framework additionally offers cross-platform compatibility, rendering it a versatile solution for various device environments.

For example, swiping motions can facilitate navigation through clothing catalogs, whereas pinching motions allow users to choose or modify garment sizes. MediaPipe's gesture recognition framework incorporates pre-processing steps for noise elimination and normalization, guaranteeing reliable performance in changing settings.



Fig. 3. hand gestures detection.

C. Emotion and Eye Tracking:

Real-time emotion recognition analyzes facial expressions to identify the user's emotions, such as happiness, sadness, or apathy. Gaze tracking identifies the areas of the screen that the user focuses on, providing valuable information about their interaction habits. These features allow the system to modify its interface in real-time, offering tailored suggestions and improving overall user experience. For instance, when a user expresses discontent with a particular piece of clothing, the system may recommend different alternatives. Likewise, gaze tracking information can pinpoint favored zones of attention, assisting designers in refining the arrangement for improved interaction.

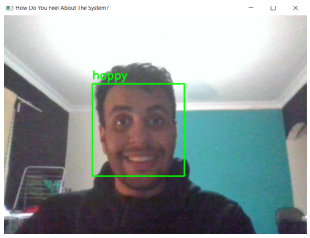


Fig. 4. Bluetooth Login .

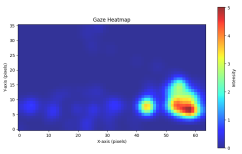


Fig. 5. Gaze Tracking .

D. YOLO for Identifying Objects

YOLO (You Only Look Once) is a widely used real-time object detection system that effectively recognizes and identifies clothing worn by individuals. Through the examination of images or video streams, YOLO is capable of identifying different types of clothing, including shirts, pants, footwear, and accessories. The system offers a confidence measure for every identification, reflecting the precision of the detection. For instance, YOLO could recognize a jacket with a 90confidence score, which indicates strong certainty that the identified object is actually a jacket. This technology can be applied in numerous settings, including virtual fitting experiences, fashion suggestions, and automated stock control in retail.YOLO is utilized to detect and identify clothing worn by the user. For example, it can recognize a jacket with a 90

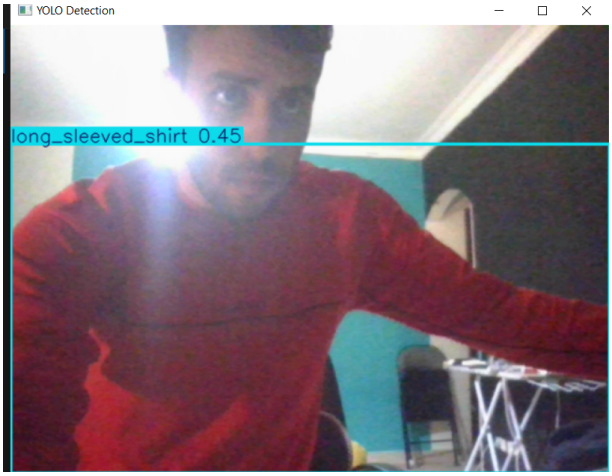


Fig. 6. YOLO Detection .

E. AR-Driven Virtual Clothing Fitting

Using TUIO to display an AR model enables users to interact with augmented reality in real-time. This module leverages TUIO protocols to track multi-touch inputs and gestures, allowing users to manipulate and view virtual models seamlessly. Through AR overlays, users can explore interactive designs or 3D representations with precision. The system integrates dynamic motion tracking and lighting adjustments, ensuring a highly immersive experience that provides a realistic preview of the model from various angles. This approach enhances user engagement, making it an ideal tool for education, design visualization, and interactive demonstrations.

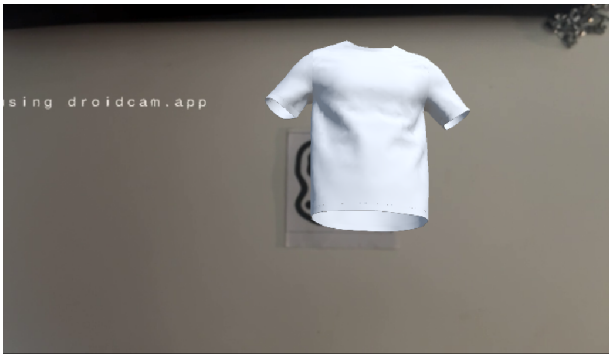


Fig. 7. AR 3D Model.

F. Bluetooth or Face Login Technologies

Bluetooth and Face Recognition for Authentication: Bluetooth and face recognition provide secure and convenient ways to log in to systems. For Bluetooth, the system identifies the user's paired device, allowing for seamless access when the device is within range. With face recognition, the system uses advanced algorithms to scan and authenticate the user's facial features, offering a hands-free login experience. Both methods enhance security while simplifying the authentication process, ensuring that only authorized individuals can access the system. The integration of these technologies promotes a smoother and more efficient user experience, making login quicker and more intuitive.

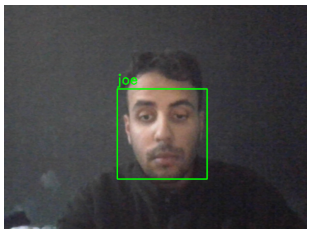


Fig. 8. face Login .

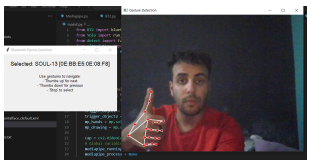


Fig. 9. Bluetooth Login .

EXPERIMENTS

To evaluate the system's efficiency, a series of experiments was conducted involving 10 participants. The experiments focused on these elements: Precision in Gesture Identification: MediaPipe was assessed for its ability to recognize specific hand gestures. The system achieved an average accuracy of 95 across all participants. Reliability of Object Detection: The assessment of YOLO's effectiveness included participants

donning various clothing pieces. The mean detection accuracy stood at 92, upholding a confidence threshold of 80AR Visualization Standard: Participants evaluated the realism and reactivity of the virtual fitting utilizing AR technology on a scale from 1 to 10, resulting in an average score of 8.5. Performance in Emotion and Eye Tracking: Emotion detection was validated by comparing it against self-reported user emotions, achieving an accuracy of 87. Gaze tracking successfully identified areas of user attention with 90 precision. Simulation of Real-World Deployment

A regulated experiment replicating an actual online shopping situation was carried out. Participants were assigned to choose and virtually try on clothing, then proceed to actual purchase and delivery. The satisfaction with the fit after receiving the physical items aligned with the virtual fitting in 85 of instances, showcasing the system's effectiveness in influencing purchase choices.

EVALUATION

The experimental results confirm the system's ability to provide an engaging and user-friendly virtual clothing fitting experience, showcasing its efficiency across various assessment standards. Essential performance indicators consist of gesture recognition accuracy, AR realism, and how emotion and gaze tracking influence personalization and engagement.

The Scenario to test the system 1. System instructions are given to the user 2. Use TUI marker to navigate to LongSleeve Category and select the white color 3. use mediapipe to do the same task

User	Age	TUIO Completion Time	Mediapipe Completion Time
Mostafa Ahmed	24	17s	35s
Ahmed	55	14.78s	72s
Sara	12	13.87s	46.53s
Ali Sameh	20	10s	47s
Hamza	8	15s	78s
Essam	53	12s	120s
Average	-	**13.775s**	**66.42s**
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TABLE I
TUIO AND MEDIAPIPE COMPLETION TIMES

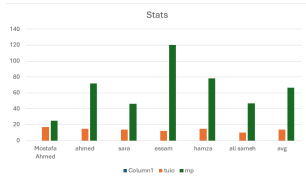


Fig. 10. Bar Chart

Results: Tuio system is much better in speed, accuracy and user comfort
Rank:
1.TUIO 2.Mediapipe
Learnings:
1. Task Completion Time:
Using MediaPipe gestures and skeleton detection, the average task completion time was 47.84 seconds. With TUIO markers, the average task completion time was significantly lower at 15.22 seconds.
2. Effectiveness:
TUIO markers proved to be a faster and more efficient method for completing tasks in the virtual fitting room system. MediaPipe provided a more interactive experience, leveraging advanced gesture and skeleton detection technologies.
Problems:
1. TUIO Markers:
1-Accuracy Issue: TUIO marker rotation was not consistently accurate, leading to occasional misalignments and errors.
2. MediaPipe:
1-Slow Response: MediaPipe's response time was noticeably slower, resulting in delays during interactions. 2-User Discomfort: The gesture-based method caused physical discomfort, such as arm pain, particularly during extended use. 3-Complexity of Instructions: The instructions for using the MediaPipe method were hard for users to remember and understand, impacting the overall usability and experience.

DISCUSSION

The incorporation of cutting-edge technologies like AR, AI, and 3D modeling into virtual fitting rooms represents a major progression in the fashion retail sector. These systems enhance the shopping experience while also tackling significant problems like elevated return rates and ecological effects. By providing tailored and engaging try-on experiences, VFRs boost user contentment and confidence in their purchases. Nonetheless, obstacles like data security, scalability, and device compatibility continue to hinder wider adoption.

The results shown in this paper highlight the capability of VFR systems to transform e-commerce. For example, employing MediaPipe for recognizing gestures and tracking emotions boosts interactivity and personalization, whereas AR-based virtual clothing fitting offers an authentic preview of outfits. These attributes greatly diminish the uncertainty linked to online shopping, thus promoting trust and loyalty among customers.

In spite of these advantages, ethical issues related to data privacy and the environmental effects of growing digital consumption require additional exploration. The implementation of blockchain technology for secure data handling and the inclusion of sustainability metrics in VFR systems are encouraging avenues for future investigation.

CONCLUSION

Virtual fitting rooms signify a transformation change in the e-commerce realm, providing creative answers to enduring issues in online fashion sales. Utilizing AR, AI, and 3D modeling, these platforms offer engaging and tailored shopping experiences that improve customer satisfaction and lower return rates. This paper emphasizes the technological progress and enhancements in usability that support contemporary VFR systems, along with the challenges that need to be tackled to guarantee their broad acceptance.

As we look to the future, the prospects of incorporating new technologies like haptic feedback, virtual reality, and cutting-edge AI systems present thrilling chances to enhance virtual fitting rooms even more. Tackling scalability challenges and guaranteeing device compatibility will be essential for broadening access among various user demographics. Ethical issues, especially those related to privacy and data protection, should continue to be a central focus in order to foster consumer confidence. By persistently innovating and tackling these issues, VFR systems can lead to a more sustainable, engaging, and inclusive future in online commerce.

Future Work

Future studies ought to concentrate on a few essential domains to enhance the functionality and acceptance of virtual fitting room (VFR) technologies. A promising avenue consists of merging haptic feedback with virtual reality (VR) technologies to develop a more immersive and tactile shopping experience. Haptic feedback, for instance, might enable users to experience the texture or weight of fabrics in a virtual environment, greatly boosting realism. Likewise, progress in AI and machine learning might result in more advanced recommendation systems that can evaluate intricate user preferences, purchase history, and present trends to provide exceptionally tailored suggestions.

Another important aspect is tackling the scalability of these systems. Creating lightweight algorithms and cloud-based solutions can guarantee compatibility across various devices, ranging from smartphones to advanced VR headsets, allowing VFRs to reach a wider audience. Additionally, improving cross-platform interoperability will be vital for enabling smooth integration with current retail systems.

Ethical and privacy issues also require additional examination. It is crucial to guarantee safe data storage and utilization by implementing strong encryption and adhering to international privacy regulations. Blockchain technology may serve as a feasible solution, offering transparency and traceability in data management while enhancing user trust. The realm of sustainability in fashion presents another opportunity for VFR systems to create a notable influence. Upcoming systems might integrate environmental indicators, like carbon emissions or water consumption data, into the user interface, enabling users to make more environmentally friendly decisions. Furthermore, partnerships with fashion designers and retailers might investigate methods to enhance garment designs for both digital and physical environments, minimizing waste and encouraging sustainable production methods.

Ultimately, investigating social and gratification components in VFRs may further boost user involvement. Elements such as virtual fashion events, live collaboration with friends, or loyalty incentives for trying on clothes online could mimic and potentially exceed the social elements of shopping in physical stores

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